

Rishabh Bhattarai

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EDUCATION

University of Michigan - College of Engineering | Ann Arbor, MI

April 2027

B.Sc Computer Engineering | GPA: 3.7

Relevant Coursework: VLSI Design, Digital Circuits, Logic Design, Computer Architecture, Data Structures & Algorithms

WORK EXPERIENCE

Firmware Engineer Co-op | Sierra Nevada Corporation | Remote

December 2025 - Present

- Supporting FPGA development workflows using VHDL.

Systems Engineer Intern | Sierra Nevada Corporation | Englewood, CO

May 2025 - December 2025

- Developed a reusable SysML-based systems engineering meta-model used across 4+ engineering teams, improving hardware/software interface traceability and reducing design rework by ~25%.
- Integrated an Arduino-powered robotic arm with SNC internal tools using C++; documenting end-to-end workflows, interface mappings, and connectivity specifications to enable scalable future third-party tool integrations.
- Collaborated with 10+ cross-functional engineers on architecture development, lifecycle modeling, and modernization of legacy processes, contributing to multiple requirement reviews for aerospace and defense systems.
- Contributed to model validation in Cameo Enterprise Architecture, improving traceability and ensuring alignment with stakeholder requirements in defense/aerospace projects through detailed reviews of system behaviors.

SharePoint Developer | MSU Facility for Rare Isotope Beams | East Lansing, MI

May 2024 - January 2025

- Designed 4+ Microsoft SharePoint Online (SPO) applications to streamline and automate routine business workflows.
- Automated 10+ business processes using Microsoft Power Automate, reducing manual work by an estimated 30%.
- Tested and deployed SPO applications in production environments, following Agile development methodologies.
- Provided timely technical support through ticketing systems for 500+ laboratory engineers, diagnosing software issues, resolving system errors, troubleshooting site deployments, and improving overall productivity.

PROJECTS

8-Bit Dual-Mode Ripple Carry Adder (Cadence Virtuoso, Maestro, CMOS/VLSI Design)

December 2025

- Designed and implemented a complete 8-bit dual-mode ripple carry adder at the transistor level, including full adders, registers, multiplexers, and inverter networks in Cadence Virtuoso.
- Built a full Spectre-based simulation and verification environment with clocked testbenches, vector-driven inputs, and unified power rails to validate functional correctness, timing, and power constraints.
- Performed transistor sizing studies to create high-performance and low-power design variants, analyzing delay, dynamic power, and leakage under representative switching activity.
- Used pipelining with input/output registers to constrain the ripple-carry critical path and improve timing margins.
- Debugged timing and functional issues through waveform analysis and iterative schematic refinement.

Sequential Four-Function Calculator (Verilog RTL, ModelSim, Altera DE2 FPGA)

April 2025

- Designed and implemented a signed 11-bit sequential calculator on an Altera DE2-115 FPGA supporting addition, subtraction, multiplication, and division through a shared datapath and finite state machine-based controller.
- Implemented signed sequential multiplication and iterative division with overflow and divide-by-zero handling.
- Developed comprehensive ModelSim testbenches to verify correctness, state sequencing, and edge cases.

Rocket Apogee Detection System | MASA (C++, C, Python)

March 2025

- Built a simulation and software pipeline to detect a rocket's peak altitude using sensor data and noise modeling.
- Developed physics-based signal processing logic to improve accuracy and reliability during flight tests.

TECHNICAL SKILLS

Skills: CMOS Digital Design, Timing Analysis, Cadence Virtuoso/Maestro, ModelSim, FPGA Design, Systems Engineering

Programming: C/C++, Python, Tcl, JavaScript, SQL, ARM Assembly, Linux, Git, MATLAB, SysML, Verilog/VHDL